

SUPERVISED LEARNING HANDS-ON PROJECT

Customer Churn Prediction – Messy Real-World Dataset

Business Context

A subscription-based company is experiencing customer churn. Management wants to identify customers who are likely to leave so that the retention team can intervene before the customer churns.

Your role

You are a Junior Data Scientist. You have been given a raw customer dataset containing demographic, usage, billing, service, and satisfaction information. The dataset intentionally contains common real-world data-quality issues.

Business Objective

Build a supervised machine learning solution that predicts whether a customer is likely to churn (Churn = Yes/No) and provide evidence that the model is reliable on unseen data.

Dataset

File: Supervised_Learning_Messy_Customer_Churn.csv

Rows: 184 including intentional duplicate records

Target: Churn

Suggested End-to-End Tasks

1. Understand the problem

- Identify the target variable.
- Identify numerical, categorical, and identifier columns.
- Explain whether this is a classification or regression problem.

2. Import and initial inspection

- Load the CSV using pandas.
- Inspect shape, data types, sample records, descriptive statistics, and target distribution.
- Identify suspicious values and potential data-quality problems.

3. Data cleaning

Investigate and handle:

- Missing values
- Duplicate records
- Invalid numerical values
- Outliers
- Inconsistent categorical labels/capitalization/whitespace
- Placeholder values such as N/A, NA, and ?
- Impossible values such as negative income, negative usage, or unrealistic age

Document the decisions you make rather than blindly deleting rows.

4. Exploratory Data Analysis

Investigate questions such as:

- Does churn differ by contract type?
- Is churn related to satisfaction?
- Does tenure appear related to churn?
- Does monthly charge relate to churn?
- Which customer characteristics appear associated with higher churn?

Use appropriate visualizations and explain the business meaning of your observations.

5. Feature engineering and preprocessing

- Decide whether Customer_ID should be used as a feature.

- Encode categorical variables.
- Decide how missing values should be treated.
- Apply feature scaling where appropriate.
- Split the data into training and testing sets without leaking information from the test set.

6. Build multiple supervised learning models

At minimum, implement:

- Logistic Regression
- KNN
- Decision Tree
- Random Forest

Optional extension:

- AdaBoost or another boosting model

7. Model evaluation

For classification, calculate and interpret:

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1-score

Do not rely only on accuracy. Explain which metric is important for the business problem and why.

8. Cross-validation

- Apply cross-validation to compare model stability.
- Compare cross-validation performance with test performance.

- Identify signs of overfitting or underfitting.

9. Model comparison

Create a comparison table containing:

Model | Accuracy | Precision | Recall | F1 | CV Score

Discuss the trade-offs between the models. Do not choose a model only because it has the highest training score.

10. Final model

Select a final model based on the business objective and evidence from validation/testing.

Explain:

- Why this model was selected
- Which features appear important
- What types of errors the model makes
- How the business could use the predictions

11. Business recommendations

Prepare 3–5 actionable observations for the retention team.

Example output questions:

- Which customer segment appears most associated with churn?
- What customer characteristics should the retention team monitor?
- How should false negatives and false positives be treated?
- How could the model be improved with additional data?

Expected Deliverables

1. Jupyter Notebook (.ipynb)
2. Cleaned dataset (.csv)

3. Model comparison table
4. Confusion matrix for the selected model
5. 3–5 business insights
6. Short conclusion explaining the final model and its limitations

Important Training Rules

- Do not delete records simply because they look unusual; investigate first.
- Do not preprocess the entire dataset before the train/test split in a way that causes data leakage.
- Do not use Customer_ID as a predictive feature merely because it is present.
- Explain every major cleaning and modeling decision.
- Compare multiple models before selecting the final approach.